

1.1 Glossary

Table 1.1.1 shows the terms related to the operation of IPL explained in this application note.

Table 1.1.1 The term for IPL operation

Term	Description
Initial Program Loader (IPL)	A loader program that launches an FSP application for the RZ/A3 environment. It is called from the BootROM program to initialize the memory and the peripheral I/O modules installed in boards, then copies the FSP application and jumps to the copy destination according to the application header. IPL is developed based on Arm Trusted Firmware and it is equivalent to BL2 of Arm Trusted Firmware. When changing the Flash memory to be used, the user is requested to create and add the initialization code according to the specifications of the Flash memory to be used, referring to this application note.
BootROM Program	The BootROM program is the program that executes first after the device starts, and it copies the next boot program from the memory associated with the boot mode to the On-chip RAM and jumps to the next boot program. IPL is premised on operation in Boot Mode 3 (1.8-V Serial Flash Memory) or Boot Mode 4 (3.3-V Serial Flash Memory) and IPL written to the Serial Flash memory is copied to the specified address (H'0001_2000) of the On-chip RAM and the boot process is executed. This program is stored in the RZ/A3's internal ROM, so it does not need to be created by the user.
FSP application	This is a program consisting of BSP and sample programs to use the functions implemented on the board.
Application Header	A header placed at the beginning of the FSP application, used by IPL to refer to the destination address or entry point address of the FSP application. Before executing IPL, the FSP application image including the application header must be written to the specified address of the Flash memory. The write destination address of the Application Header can be changed in IPL build option, and the default is H'2002_0000.

Term	Description
Serial Flash memory driver	The Serial Flash memory driver creates parameters for executing operations that comply with the specifications of the Serial Flash memory and manages the execution order of operations. IPL can implement NOR Serial Flash memory drivers and NAND Serial Flash memory drivers.
	These drivers are implemented with specifications that depend on the Serial Flash memory. If you use a Serial Flash memory other than those listed below that are supported by the standard IPL, you will need to modify the driver to match the specifications.
	RZ/A3UL Evaluation-Board-Kit (QSPI version): AT25QL128A (Serial NOR Flash memory, Product of Renesas)
EK-RZ/A3M kit:	
	MX25L25645Q (Serial NOR Flash memory, Product of Macronix)
	W25N01GV (Serial NAND Flash memory, Product of Winbond)
SPIBSC driver	
	SPIBSC driver controls SPI Multi I/O Bus Controller (SPIBSC) built into RZ/A to access the Serial Flash memory connected to the board. This driver is called by each Serial Flash memory driver. Each Serial Flash memory driver passes settings appropriate for the memory it wants to access to this driver, ensuring optimal access to the memory.
Manual Mode	
	This is the operation mode of the SPI Multi I/O Bus Controller. In this mode, data is read / written by sending Read and Write commands to the Serial Flash memory via serial communication. The default operating mode of the SPI Multi I/O Bus Controller is Manual Mode.
External Address Space Read Mode	
	This is the operation mode of the SPI Multi I/O Bus Controller. In this mode, the device can read data by performing read access to the address space where the Serial Flash memory is mapped. This mode is enabled by switching the operation mode from Manual Mode.
OctaFlash driver	
/ OctaRAM driver	

The OctaFlash driver and OctaRAM driver create parameters for executing operations that comply with the specifications of each memory and manage the execution order of operations.

These drivers are implemented with specifications that depend on each memory.

If you use the OctaFlash and OctaRAM other than those listed below that are supported by the standard IPL, you will need to modify the driver to match the specifications.

RZ/A3UL Evaluation-Board-Kit (Octal-SPI version):

MX66UW1G45G (OctaFlash, product of Macronix)

APS51208N (OctaRAM, product of AP Memory)

Octa Memory Controller driver

Octa Memory Controller driver controls Octa Memory Controller built into RZ/A to access the OctaFlash and OctaRAM connected to the board.

This driver is called by each OctaFlash driver and OctaRAM driver.

Each OctaFlash driver (or OctaRAM driver) passes settings appropriate for the memory it wants to access to this driver, ensuring optimal access to the memory.

Configuration Mode

This is the memory access mode of the Octa Memory Controller.

In this mode, data is read / written by sending Read and Write commands to the OctaFlash and OctaRAM via serial communication.

Both Configuration Mode and Memory-map Mode can be used in Octa Memory Controller without switching the operation mode.

Memory-map Mode

This is the memory access mode of the Octa Memory Controller.

By using this mode, the device can read / write data by performing read / write access to the address space where OctaFlash and OctaRAM are mapped.

Both Configuration Mode and Memory-map Mode can be used in Octa Memory Controller without switching the operation mode.